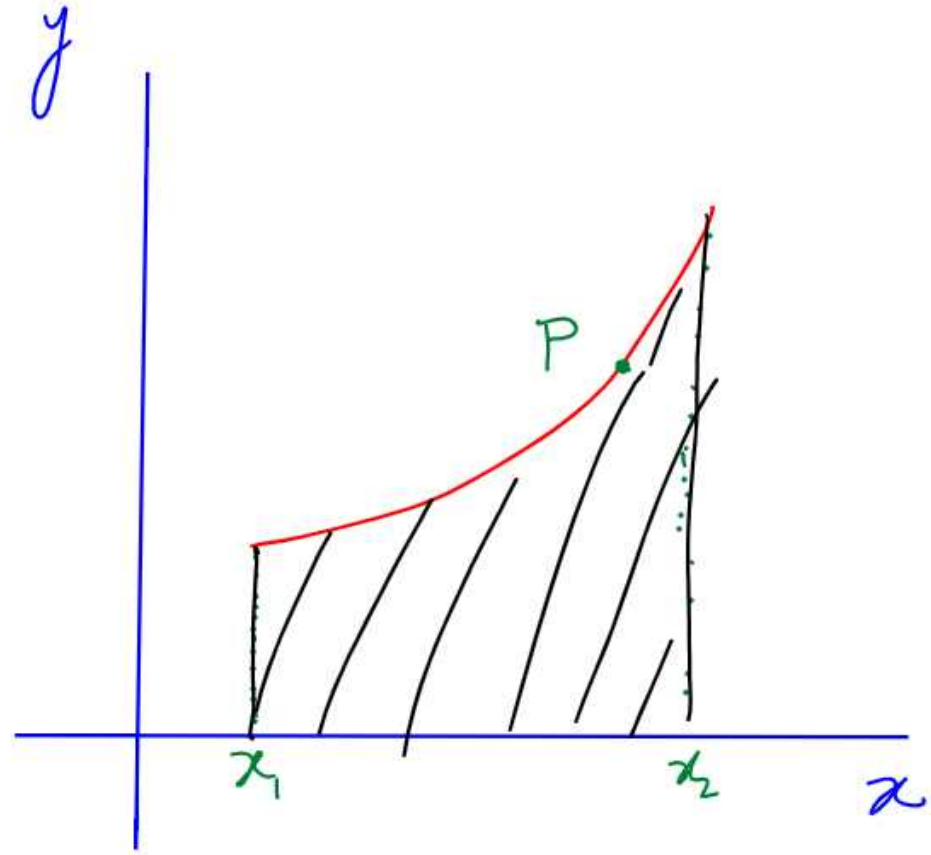
Motion in a straight line

Thu May 07 2026

$y = f(x)$

$\frac{dy}{dx}$ = slope of tangent

$\int_{x_1}^{x_2} y\,dx$ = Area under graph



$$\int F\,dt = \int \frac{dp}{dt}\,dt = \int_{P_i}^{P_f} dp = P_f - P_i = \text{Impulse}$$

$$\int i\,dt = \int \frac{dq}{dt}\cdot dt = \int_{Q_i}^{Q_f} dq = Q_f - Q_i = \Delta Q$$

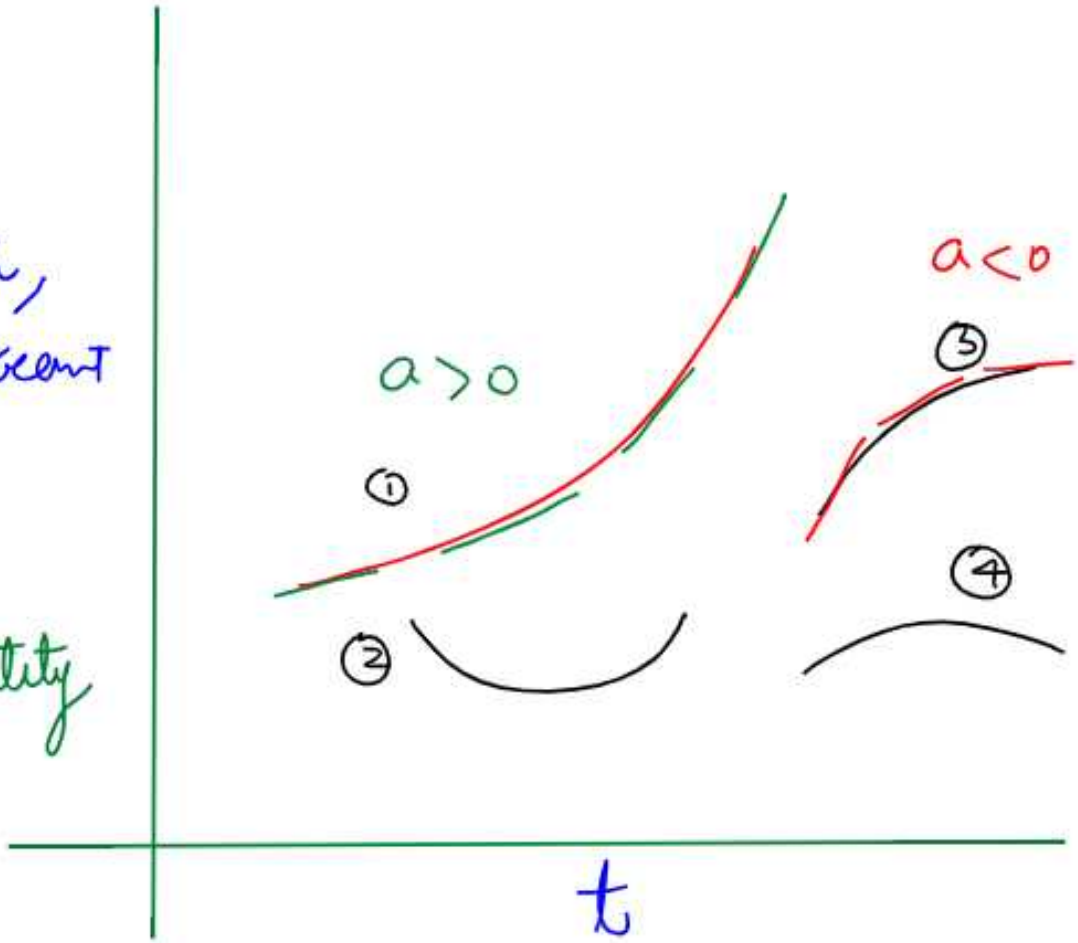
Position time / Displacement time

$\frac{ds}{dt}$, $\frac{dx}{dt}$ = velocity

position, displacement

slope → velocity

Area ⇒ No meaningful quantity



Velocity – time graph

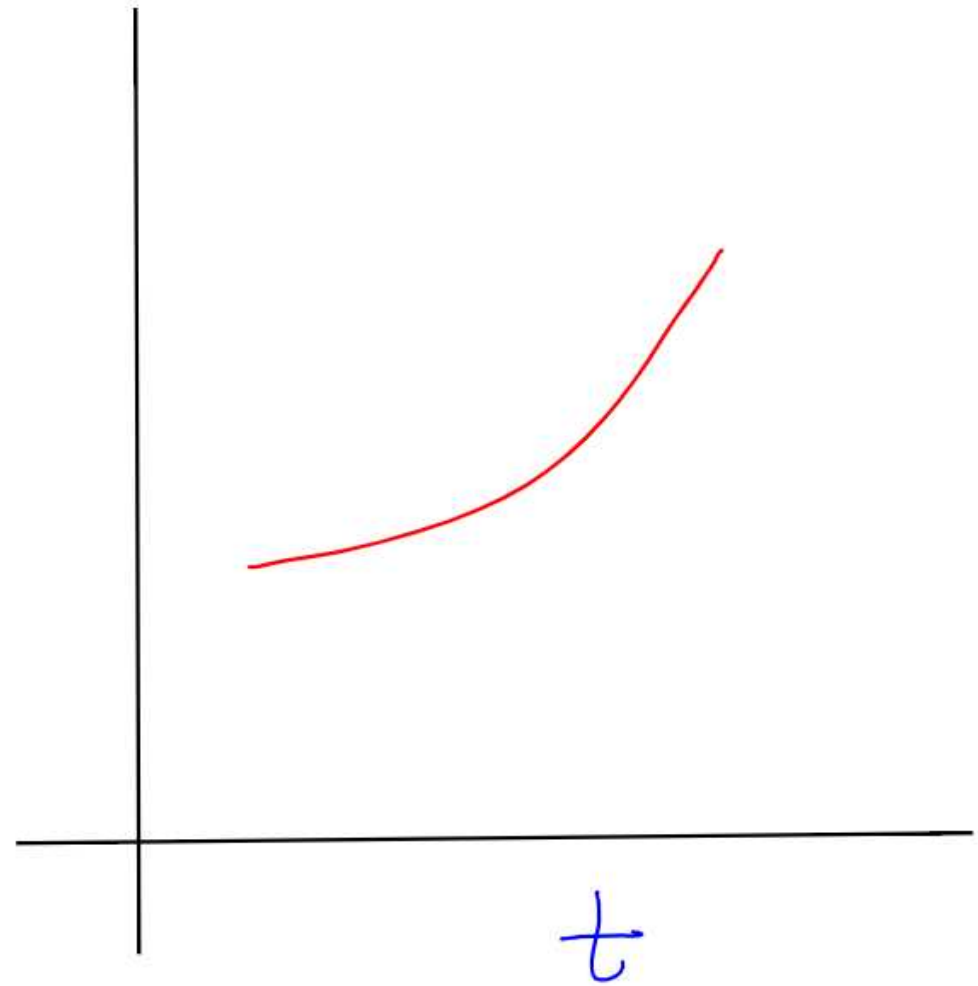
slope $= \frac{dv}{dt}$ = acceleration

$$\text{Area} = \int v\,dt$$

$$= \int \frac{ds}{dt}\,dt$$

$$= \int ds$$

$$= s = \text{displacement}$$



Acceleration - time

$$\text{Area} = \int a\,dt$$

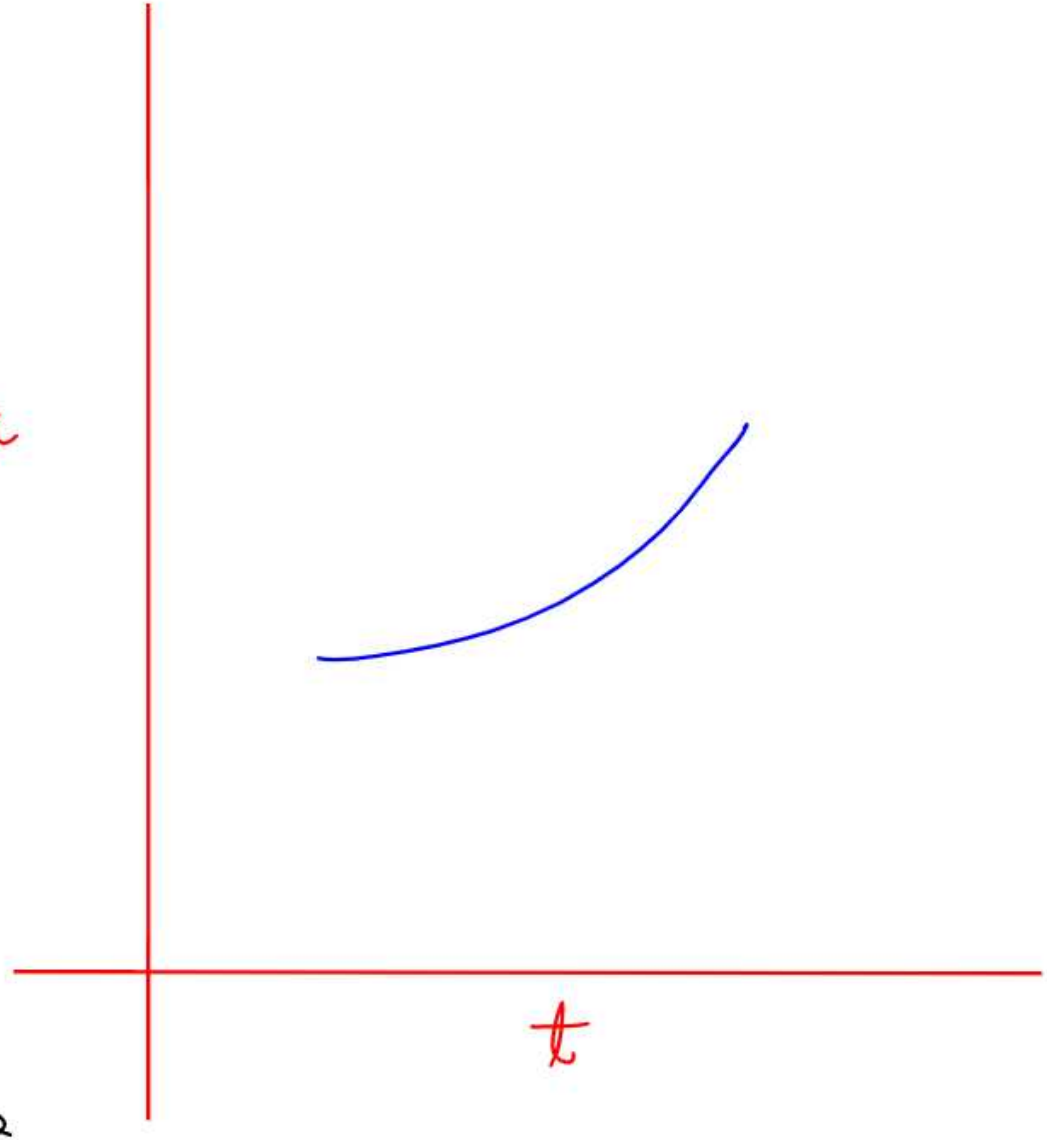
$$= \int \frac{dv}{dt} \cdot dt$$

$$= \int_u^v dv$$

$$= v\Big|_u^v$$

$$= v - u = \Delta V = \text{change in velocity}$$

$$\text{Slope} = \frac{da}{dt} = \text{jerk}$$



JEE PYQ displacement

Q7: Given below are two statements:

Statement I: Area under velocity-time graph gives the distance travelled by the body in a given time.

Statement II: Area under acceleration-time graph is equal to the change in velocity-in the given time.

In the light of given statements, choose the correct answer from the options given below.

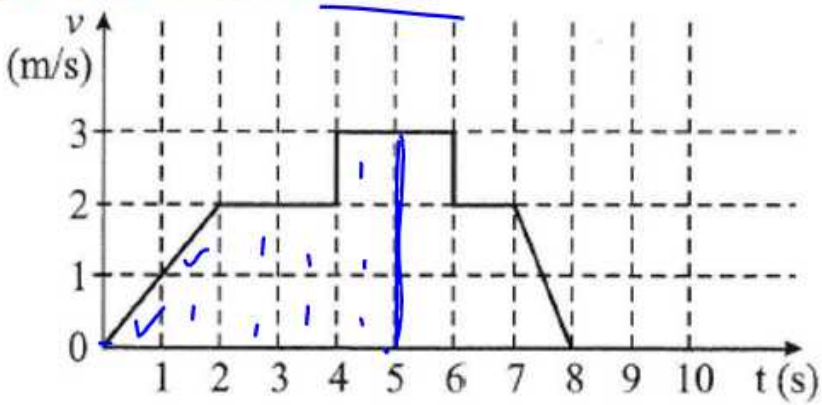
(A) Statement I is correct but Statement II is false.

(B) Statement I is incorrect but Statement II is true.

(C) Both Statement I and Statement II are true.

(D) Both Statement I and Statement II are False.

Q13: A particle starts from the origin at time t = 0 and moves along the positive x-axis. The graph of velocity with respect to time is shown in figure. What is the position of the particle at time t = 5s?





(A) 10 m

(B) 6 m

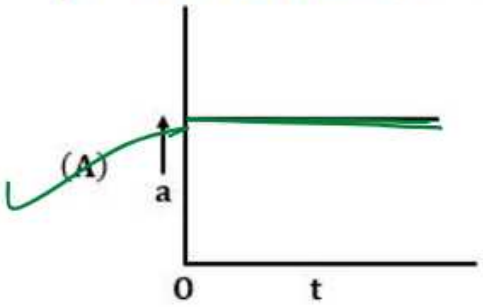
(C) 3 m

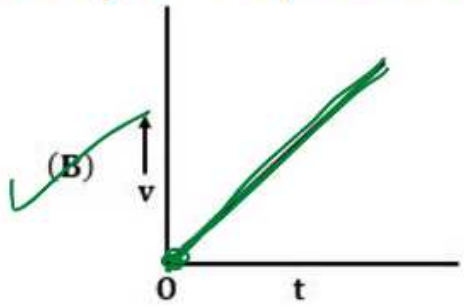
(D) 9 m

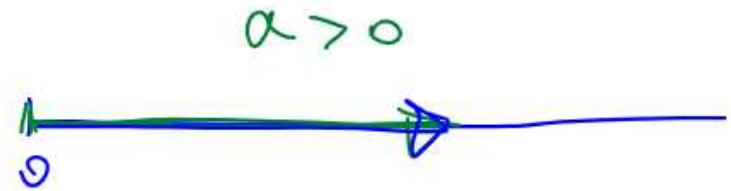
$S = 9m$

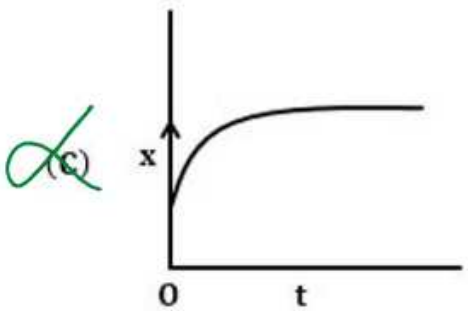
Q16: A particle starts from origin O from rest and moves with a uniform acceleration along the positive x-axis. Identify all figures that correctly represents the motion qualitatively

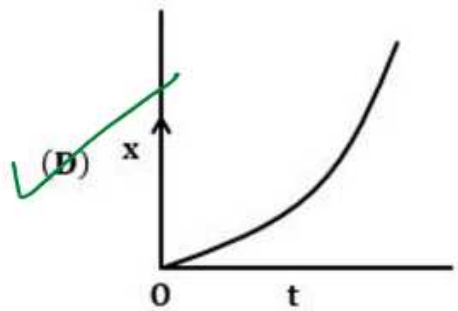
(a = acceleration, v = velocity, x = displacement, t = time)











(A) (B), (C)

(B) (A)

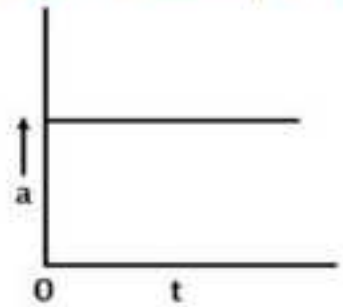
(C) (A), (B), (C)

(D) (A), (B), (D)

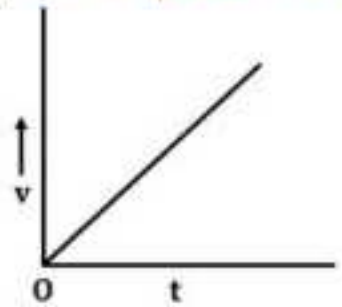
Q16: A particle starts from origin O from rest and moves with a uniform acceleration along the positive x-axis. Identify all figures that correctly represents the motion qualitatively

(a = acceleration, v = velocity, x = displacement, t = time)

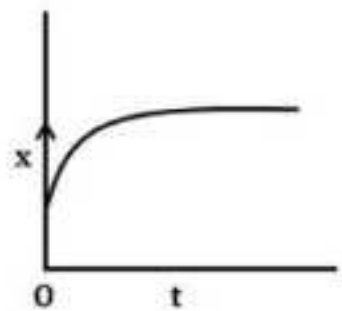
(A)



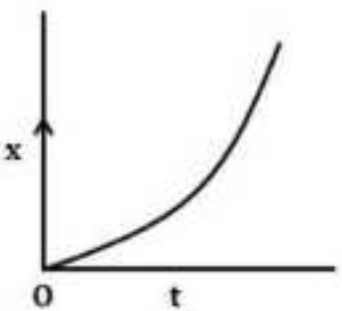
(B)



(C)



(D)



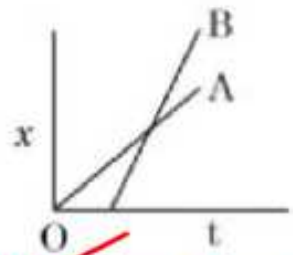
(A) (B), (C)

(B) (A)

(C) (A), (B), (C)

(D) (A), (B), (D)

Q11: The position-time graphs for two students A and B returning from the school to their homes are shown in figure.



(A) A lives closer to the school

(B) B lives closer to the school

(C) A takes lesser time to reach home

(D) A travels faster than B

(E) B travels faster than A

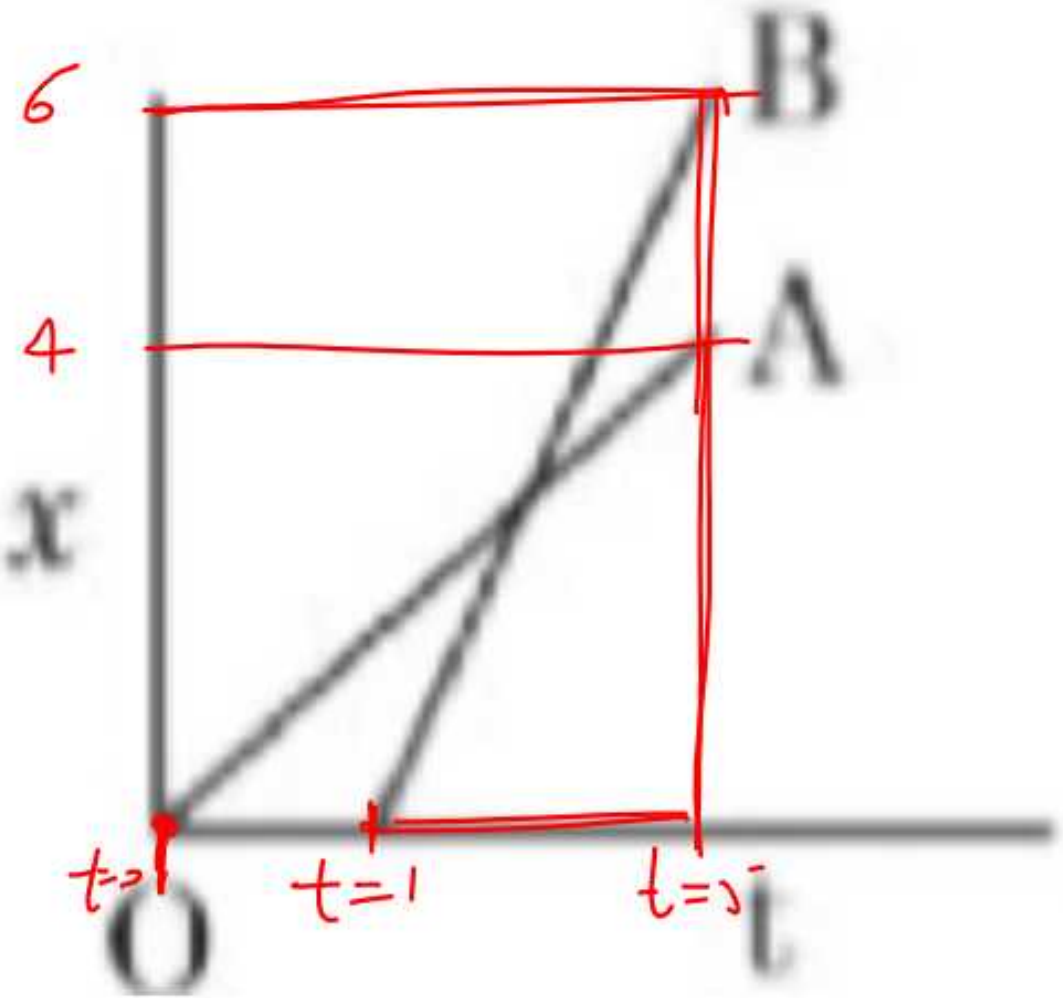
Choose the correct answer from the options given below:

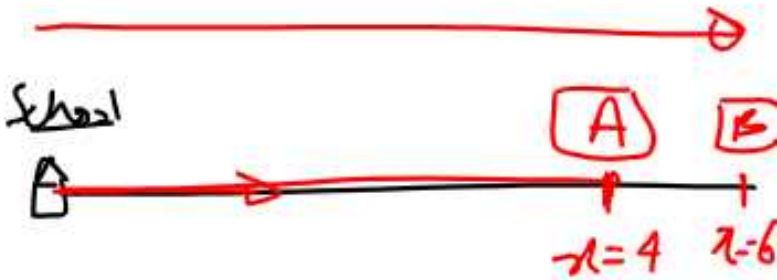
(A) (A), (C) and (E) only

(B) (A) and (E) only

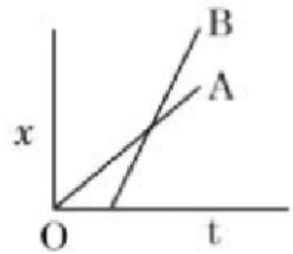
(C) (A), (C) and (D) only

(D) (B) and (E) only


6
4
x
B
A
O
t=1
t=3
t


A
B
x=4
x=6

Q11: The position-time graphs for two students A and B returning from the school to their homes are shown in figure.



(A) A lives closer to the school
(B) B lives closer to the school
(C) A takes lesser time to reach home
(D) A travels faster than B
(E) B travels faster than A
Choose the correct answer from the options given below:

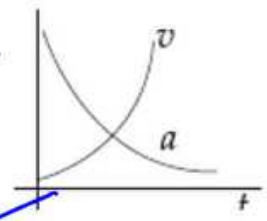
(A) (A), (C) and (E) only

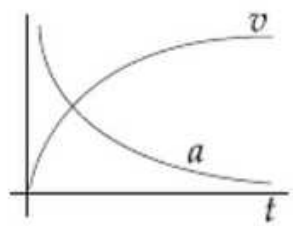
(B) (A) and (E) only

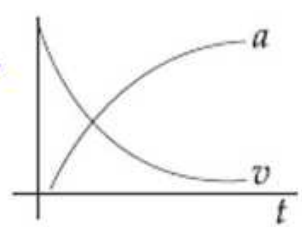
(C) (A), (C) and (D) only

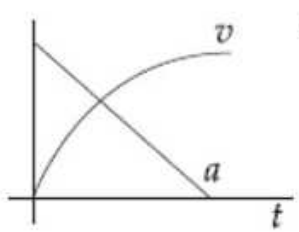
(D) (B) and (E) only

Q3: Which of the following option correctly describes the variation of the speed *v* and acceleration '*a*' of a point mass falling vertically in a viscous medium that applies a force F = −*kv*, where '*k* 'is a constant, on the body? (Graphs are schematic and not drawn to scale)

(A) 

(B) 

(C) 

(D) 

$F_{net} = mg - kv$

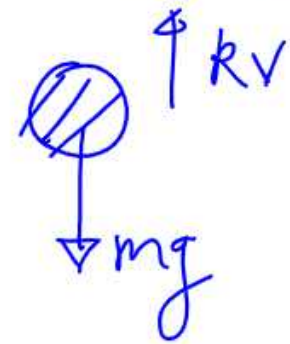
$ma = mg - kv$

$a = g - \frac{kv}{m}$

$\frac{dv}{dt} = g - \frac{kv}{m}$

$\int_0^v \frac{dv}{g - \frac{kv}{m}} = \int_0^t dt$

$\frac{\ln\left(g - \frac{kv}{m}\right)\Big|_0^v}{-k/m} = t$



$a = g - \frac{k}{m}\frac{mg}{k}\left(1 - e^{-kt/m}\right)$

$= g - g\left(1 - e^{-kt/m}\right)$

$a = ge^{-kt/m}$

$$\ln\left(g-\frac{kv}{m}\right)\Big|_0^v = -\frac{kt}{m}$$

$$\ln\left(g-\frac{kv}{m}\right)-\ln g = -\frac{kt}{m}$$

$$\ln\left(\frac{g-\frac{kv}{m}}{g}\right) = -\frac{kt}{m}$$

$$1-\frac{kv}{mg} = e^{-kt/m}$$

$$1-e^{-kt/m} = \frac{kv}{mg}$$

$$v = \frac{mg}{k}\left(1-e^{-kt/m}\right)$$

$e^{-\infty}=0$

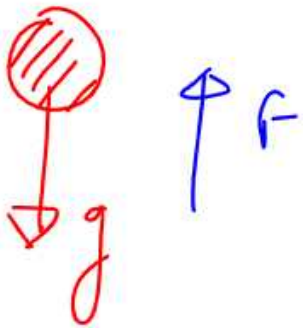
$t=0 \rightarrow v=0$

$t=\infty \rightarrow v=\frac{mg}{k}$

$6\pi\eta\gamma v$

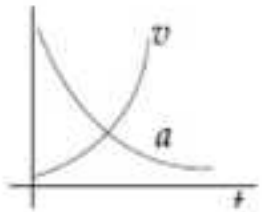
k

$F = kv$

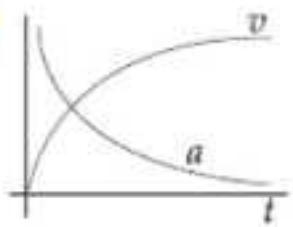


Q3: Which of the following option correctly describes the variation of the speed *v* and acceleration '*a*' of a point mass falling vertically in a viscous medium that applies a force $F = -kv$, where '*k* ' is a constant, on the body? (Graphs are schematic and not drawn to scale)

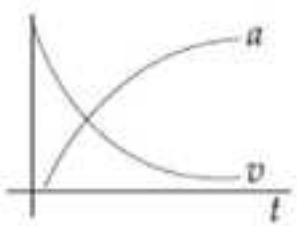
(A)



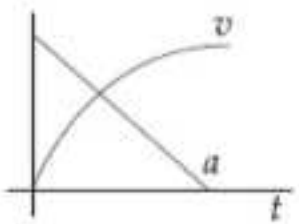
(B)



(C)

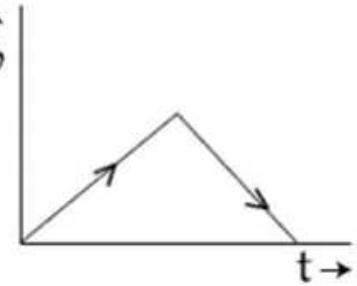


(D)

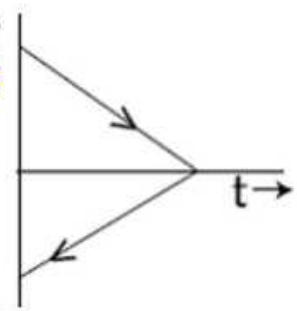


Q4: A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity vs time ?

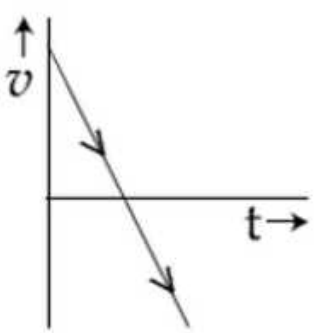
(A)



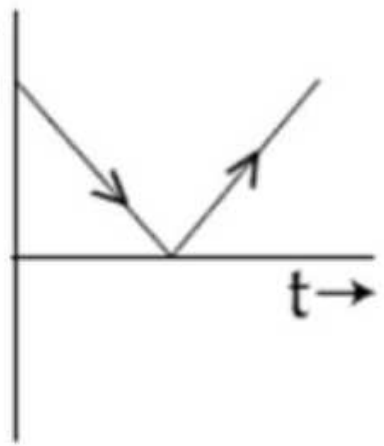
(B)

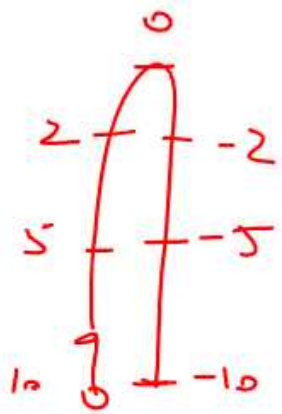


(C)

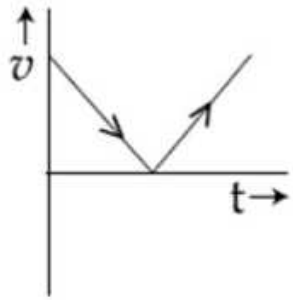


(D)





(D)



$S_{scoot} = 30 \times 15 = 450m + 7.5 \times 30 = 675$

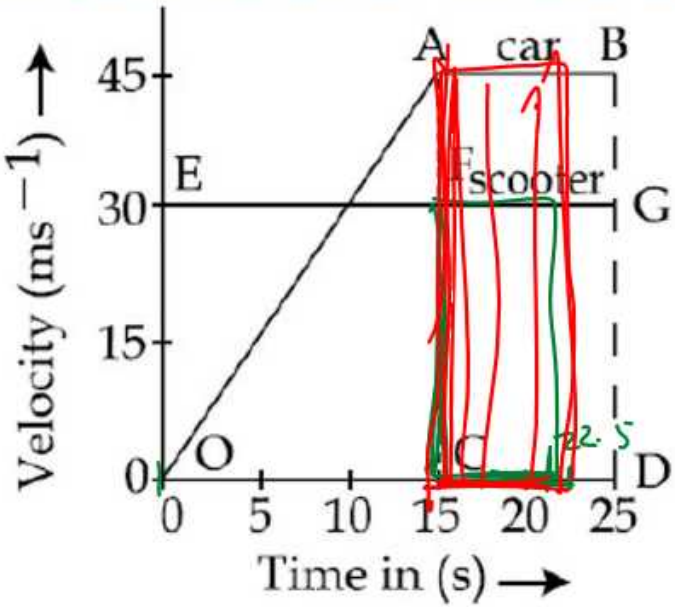
$S_{car} = \frac{1}{2} \times 15 \times 45 = \frac{675}{2} - 337.5 + 7.5 \times 45$

$= 337.5$

Q9: The velocity-time graphs of a car and a scooter are shown in the figure. (i) The difference between the distance travelled by the car and the scooter in 15 s and (ii) the time at which the car will catch up with the scooter are, respectively.

$= 675$

112.5



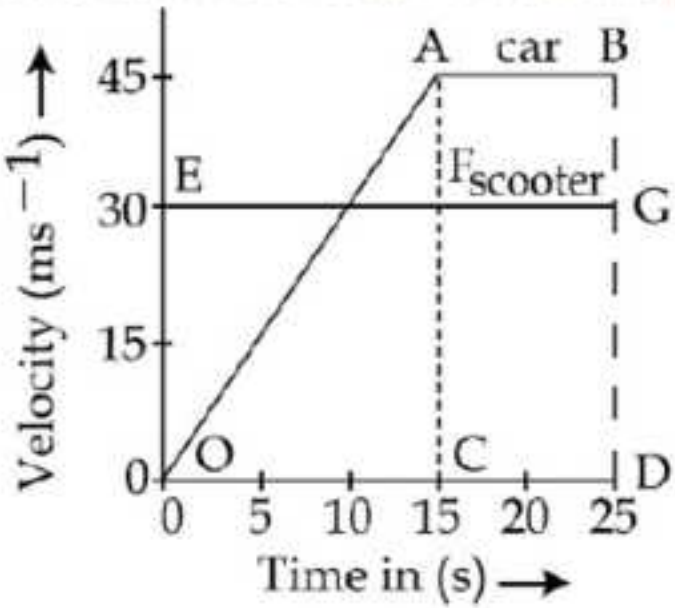
(A) 337.5 m and 25 s

(B) 225.5 m and 10 s

(C) 112.5 m and 15 s

(D) 112.5 m and 22.5 s

Q9: The velocity-time graphs of a car and a scooter are shown in the figure. (i) The difference between the distance travelled by the car and the scooter in 15 s and (ii) the time at which the car will catch up with the scooter are, respectively.



(A) 337.5 m and 25 s

(B) 225.5 m and 10 s

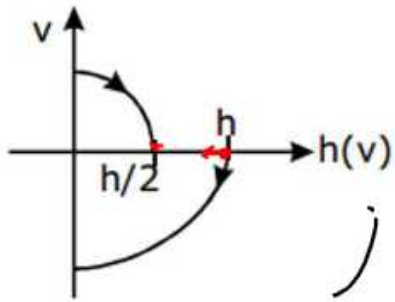
(C) 112.5 m and 15 s

(D) 112.5 m and 22.5 s

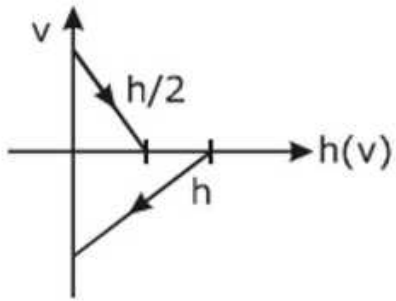
$v = \frac{dx}{dt}$ $v = \frac{ds}{dt}$

Q37: A Tennis ball is released from a height h and after freely falling on a wooden floor it rebounds and reaches height $\frac{h}{2}$. The velocity versus height of the ball during its motion may be represented graphically by: (graphs are drawn schematically and are not to scale)

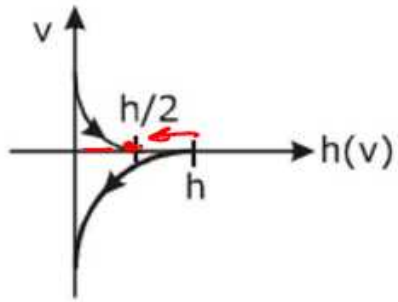
(A)



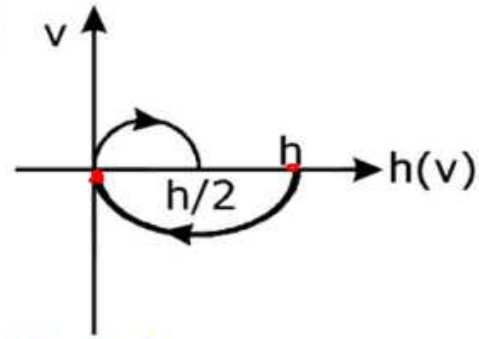
(B)



(C)

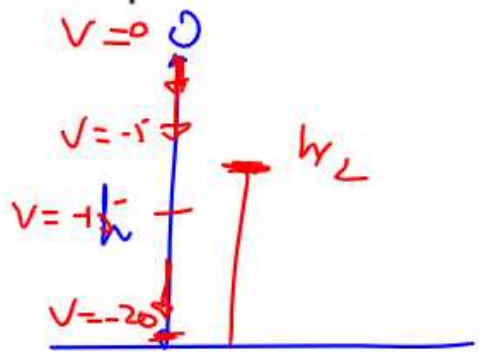


(D)

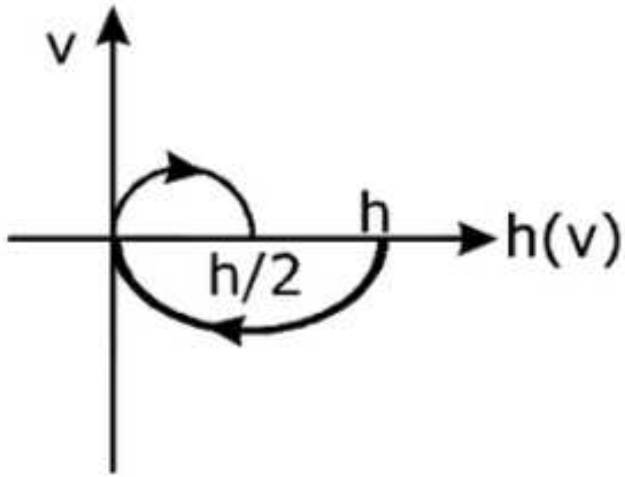


$h = ut - \frac{1}{2}gt^2$

$v^2 = u^2 - 2gh$

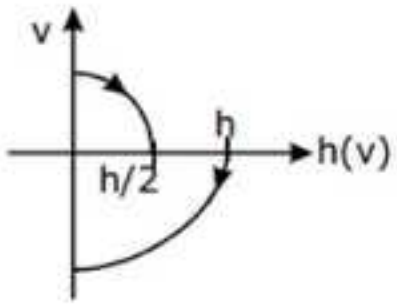


(D)

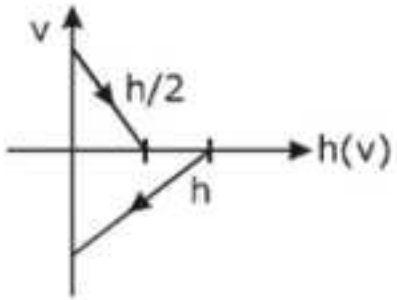


Q37: A Tennis ball is released from a height h and after freely falling on a wooden floor it rebounds and reaches height $\frac{h}{2}$. The velocity versus height of the ball during its motion may be represented graphically by: (graphs are drawn schematically and are not to scale)

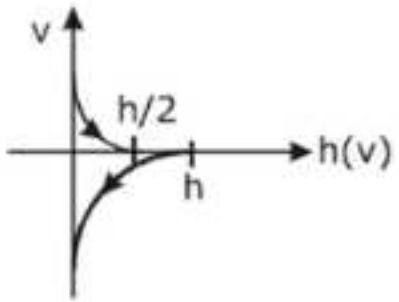
(A)



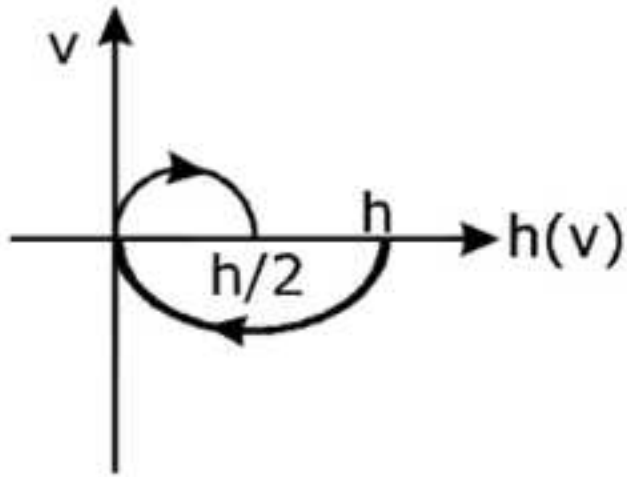
(B)



(C)



(D)



Solution:

→v-h curve will be parabolic as for motion under gratity, $\updownarrow^{+y}_{-y}$

$v^2 = u^2 \pm 2gh$

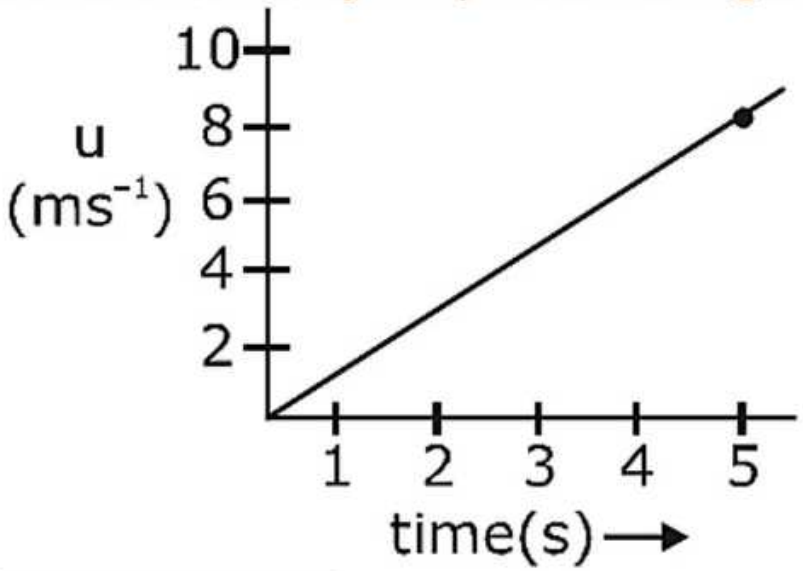
→downward velocity is negative and upward is positive.

→ when ball is coming down graph will be in IV quadrant i.e v is –ve and when going up graph will be in I quadrant i.e v is +ve.

For h = 0, v cannot be zero.

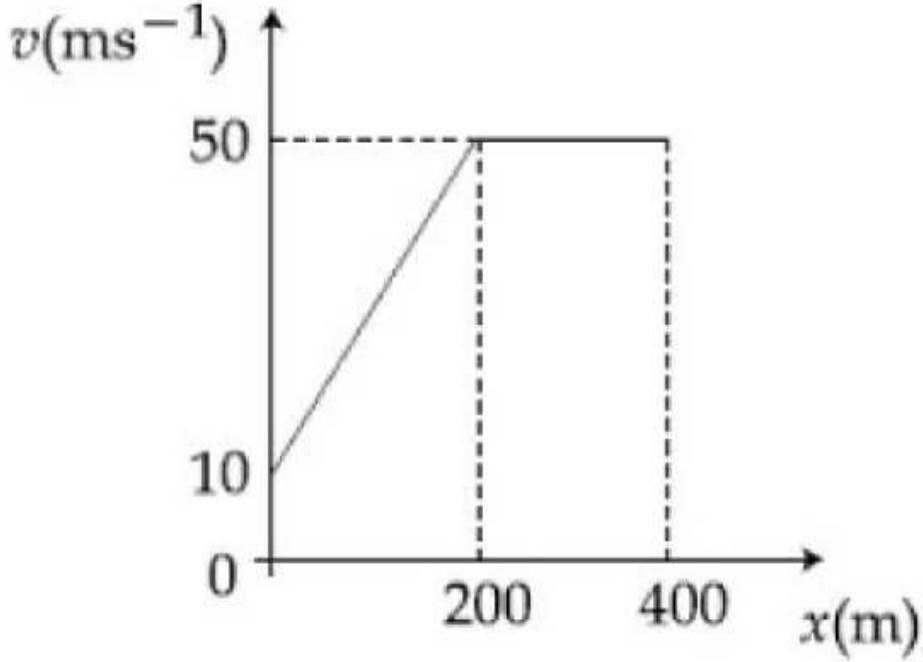
HW 2

Q45: The speed verses time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval t = 0 to t = 5 s will be_____.



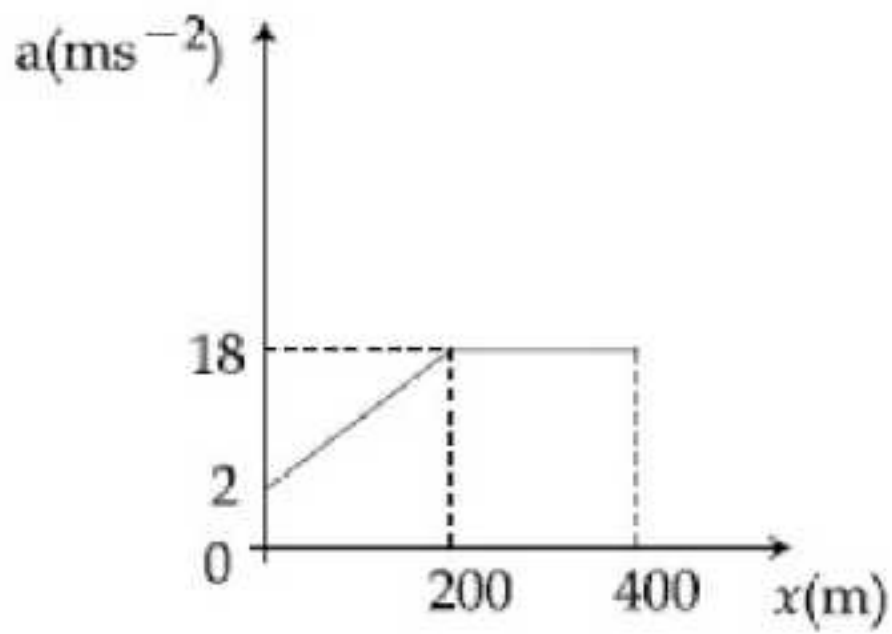
HW

Q7: The velocity-displacement graph describing the motion of a bicycle is shown in the figure.

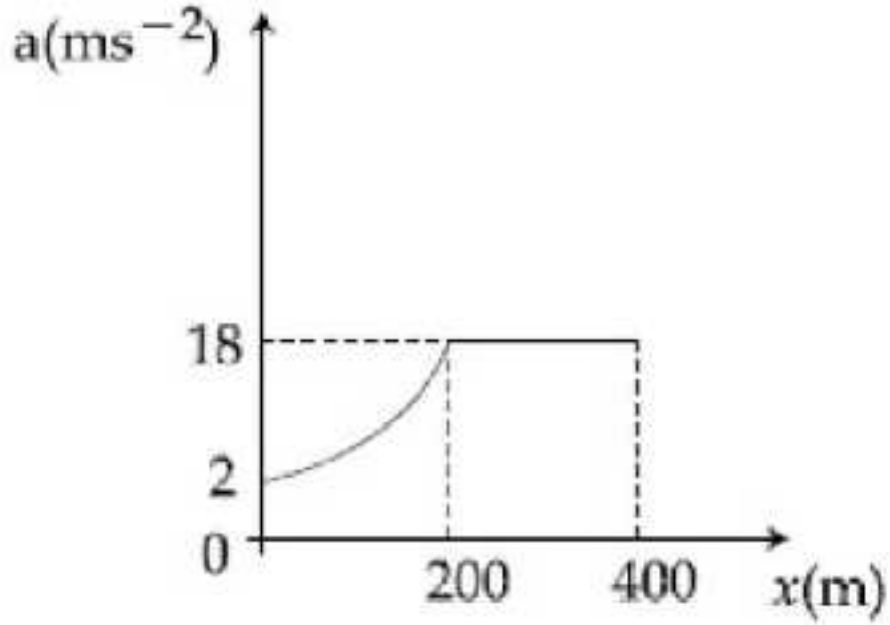


The acceleration-displacement graph of the bicycle's motion is best described by

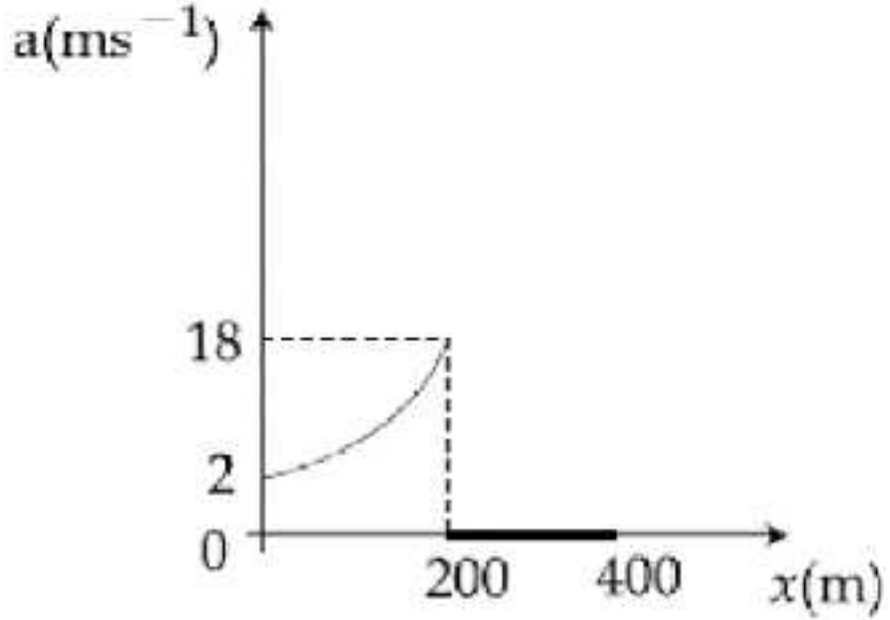
(A)



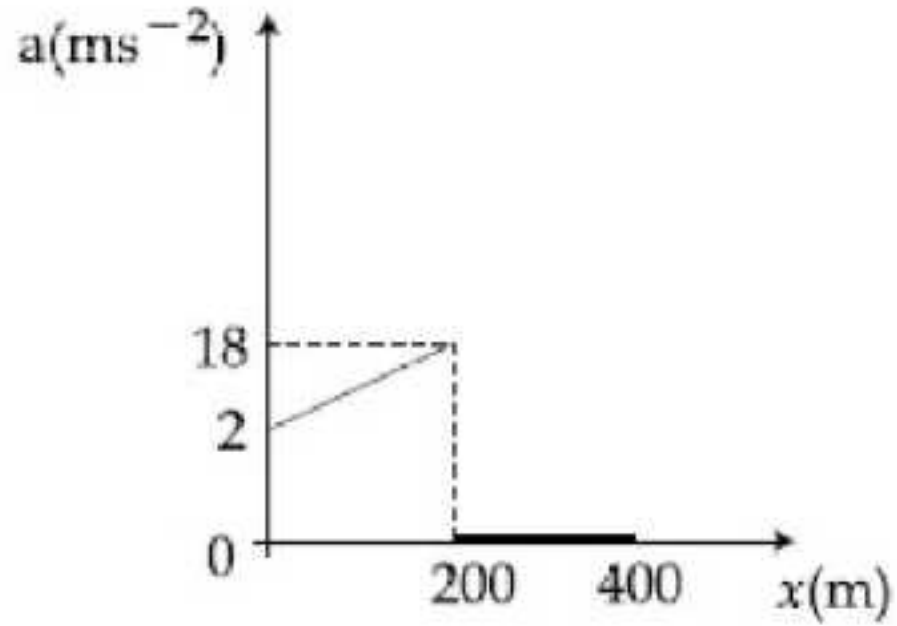
(B)



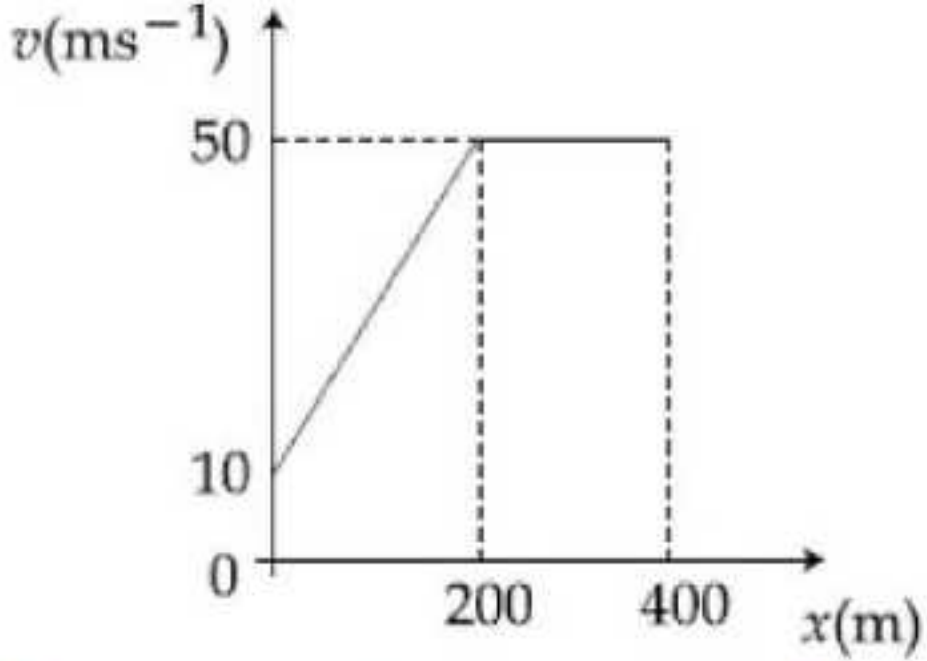
(C)



(D)

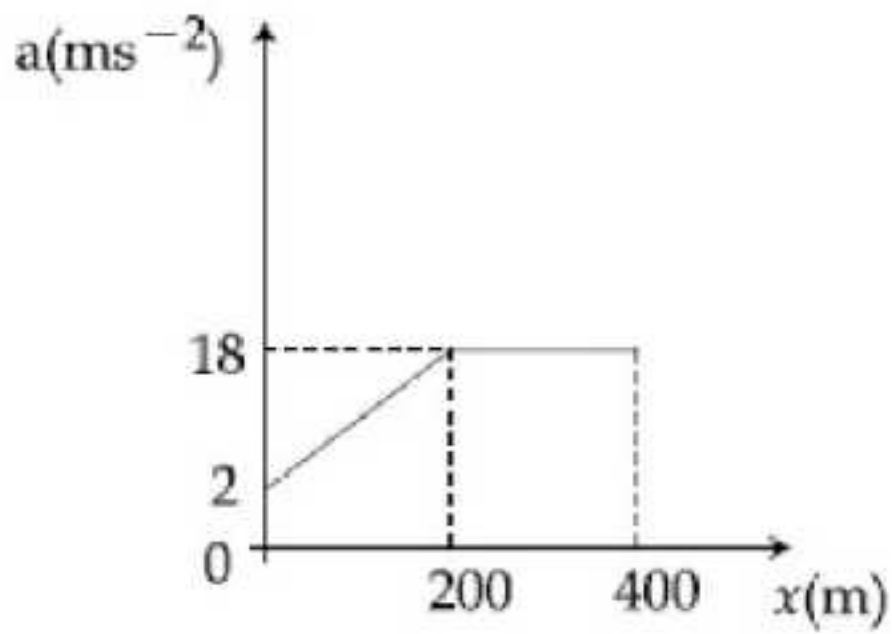


Q7: The velocity-displacement graph describing the motion of a bicycle is shown in the figure.

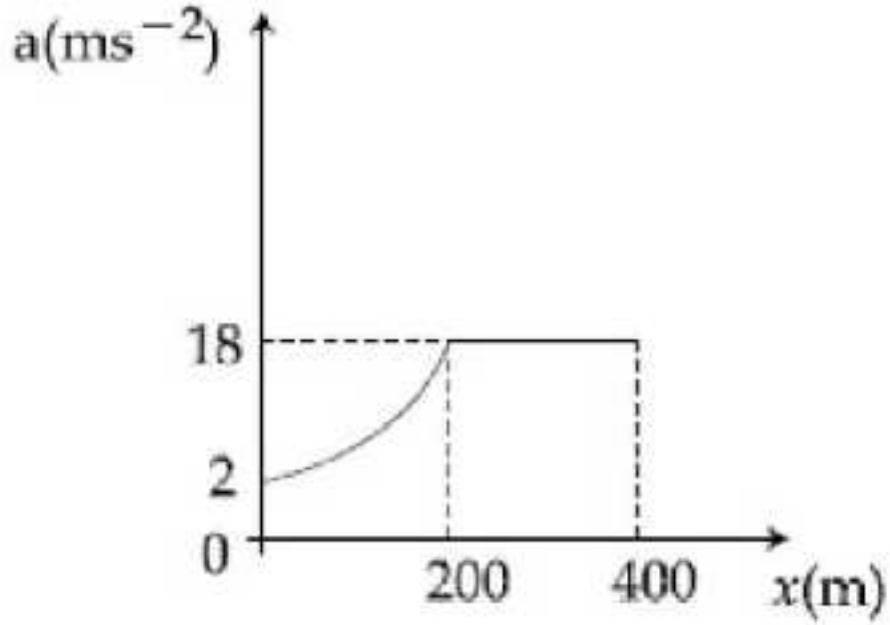


The acceleration-displacement graph of the bicycle's motion is best described by

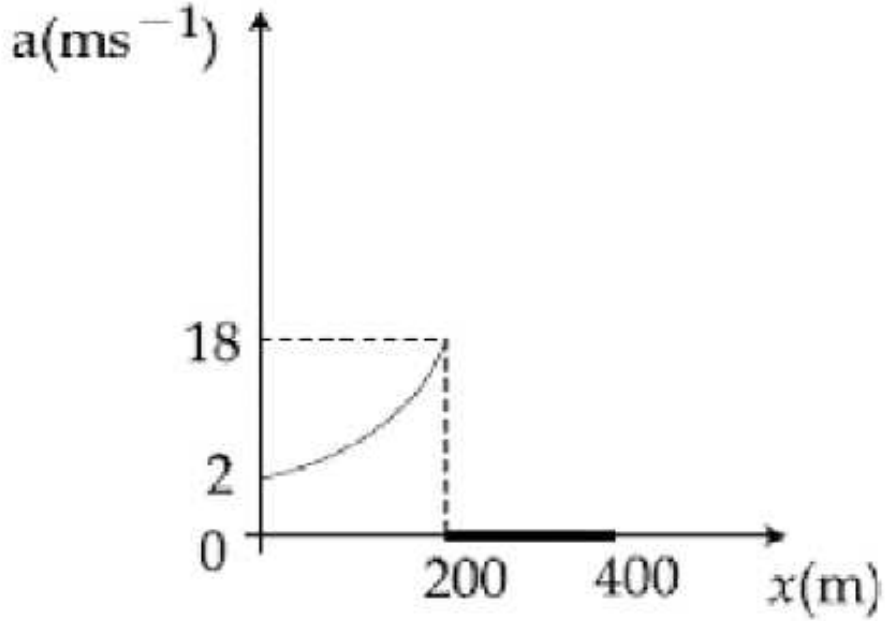
(A)



(B)



(C)



(D)

